M378K: April 10th, 2026.

Q: $\mathbb{E}[(Y-a)^2] \xrightarrow[a]{} \min$ $\qquad$ $\boxed{a^* = \mathbb{E}[Y]}$

$\mathbb{E}[|Y-a|] \xrightarrow[a]{} \min$ $\qquad$ $\boxed{a^* = \text{median of } Y}$

Maximum Likelihood Estimation.

Likelihood.

Def'n. Given a random sample $Y_1, \dots, Y_n$ from a discrete dist'n D w/ an unknown parameter θ, the likelihood function is defined as

$$L(\theta; y_1, y_2, \dots, y_n) = p^{\theta}_{Y_1, \dots, Y_n}(y_1, \dots, y_n)$$
$$= p^{\theta}_{Y_1}(y_1) \cdot p^{\theta}_{Y_2}(y_2) \cdots p^{\theta}_{Y_n}(y_n)$$
$$= p^{\theta}(y_1)\, p^{\theta}(y_2) \cdots p^{\theta}(y_n)$$

where p^{θ} is the pmf of D.

Example. Bernoulli.

$Y_1, Y_2, \dots, Y_n \sim B(p)$ $\qquad$ $\boxed{p \longleftrightarrow \theta}$

pmf of $B(p)$: $\boxed{p(y) = \begin{cases} p & \text{if } y=1 \\ 1-p & \text{if } y=0 \end{cases}}$

$$\boxed{p(y) = p^y (1-p)^{1-y}} \quad y = 0, 1$$

$$L(p; y_1, \dots, y_n) = p^{y_1}(1-p)^{1-y_1}\, p^{y_2}(1-p)^{1-y_2} \cdots p^{y_n}(1-p)^{1-y_n}$$
$$= \prod_{i=1}^{n} p^{y_i} \cdot \prod_{i=1}^{n} (1-p)^{(1-y_i)}$$
$$= p^{\sum y_i}(1-p)^{n-\sum y_i}$$

For computational reasons, take the $\ln(\cdot)$ increasing and get the log-likelihood, i.e.,

$$\ell(p; y_1, \dots, y_n) = \sum_{i=1}^{n} y_i \cdot \ln(p) + \left(n - \sum_{i=1}^{n} y_i\right) \cdot \ln(1-p) \xrightarrow[p]{} \max$$

Next, we differentiate <u>with respect to p</u>.

$$\ell'(p; y_1, \dots, y_n) = \sum_{i=1}^{n} y_i \cdot \frac{1}{p} + \left(n - \sum_{i=1}^{n} y_i\right) \cdot (-1) \cdot \frac{1}{1-p}$$

We equate this derivative to zero and solve for p.

$$\left(\sum y_i\right) \cdot \frac{1}{p} - \left(n - \sum y_i\right) \cdot \frac{1}{1-p} = 0 \quad / \cdot p(1-p)$$
$$(1-p)\sum y_i - p\left(n - \sum y_i\right) = 0$$
$$\sum y_i - \cancel{p\sum y_i} - pn + \cancel{p\sum y_i} = 0$$
$$p = \frac{\sum y_i}{n} = \bar{y}$$
$$\boxed{\hat{p}_{MLE} = \bar{Y}}$$

$\square$

Def'n. If $Y_1, \dots, Y_n$ come from a continuous dist'n D w/ pdf f^{θ}, the the likelihood f'tion is

$$L(\theta; y_1, \dots, y_n) = f^{\theta}_{Y_1, \dots, Y_n}(y_1, \dots, y_n) = f^{\theta}(y_1) \cdot f^{\theta}(y_2) \cdots f^{\theta}(y_n)$$

Example. Exponential.

$$Y_1, \dots, Y_n \sim E(\theta)$$

the pdf is $f^{\theta}(y) = \frac{1}{\theta} e^{-\frac{y}{\theta}}$, $y > 0$

$$\Rightarrow L(\theta; y_1, \dots, y_n) = \frac{1}{\theta} e^{-\frac{y_1}{\theta}} \cdot \frac{1}{\theta} e^{-\frac{y_2}{\theta}} \cdots \frac{1}{\theta} e^{-\frac{y_n}{\theta}}$$

$$= \left(\frac{1}{\theta}\right)^n e^{-\frac{1}{\theta} \sum y_i}$$

$$\Rightarrow \ell(\theta; y_1, \dots, y_n) = -n \ln(\theta) + \left(-\frac{1}{\theta} \sum y_i\right)$$

$$\Rightarrow \ell'(\theta; y_1, \dots, y_n) = -n \cdot \frac{1}{\theta} + \left(+\frac{1}{\theta^2} \sum_{i=1}^{n} y_i\right) = 0 \quad / \cdot \theta^2$$

$$-n\theta + \sum_{i=1}^{n} y_i = 0$$

$$\theta = \bar{y} \quad \Rightarrow \quad \boxed{\hat{\theta}_{MLE} = \bar{Y}}$$

$\square$